Experimental design and Markov decision processes

The following problems

- Shortest path problems.
- Optimal stopping problems.
- Reinforcement learning problems.
- Experiment design (clinical trial) problems
- Advertising.

can be all formalised as Markov decision processes.

Applications

- Robotics.
- Economics.
- Automatic control.
- Resource allocation

Contents

The n meteorologists problem

Setting

- n meteorologists.

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- At time t:

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- You must decide whether or not to commute by bike that day $a_t \in \{\texttt{bike}, \texttt{tram}\}$

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- You then store the information y_t about the weather and the meteorologists' predictions.

Utility

$$U = \sum_t r_t$$

The n meteorologists problem is simple, as:

- You always see their predictions, as well as the weather, no matter whether you bike or take the tram (full information)
- Your actions do not influence their predictions (independence events)

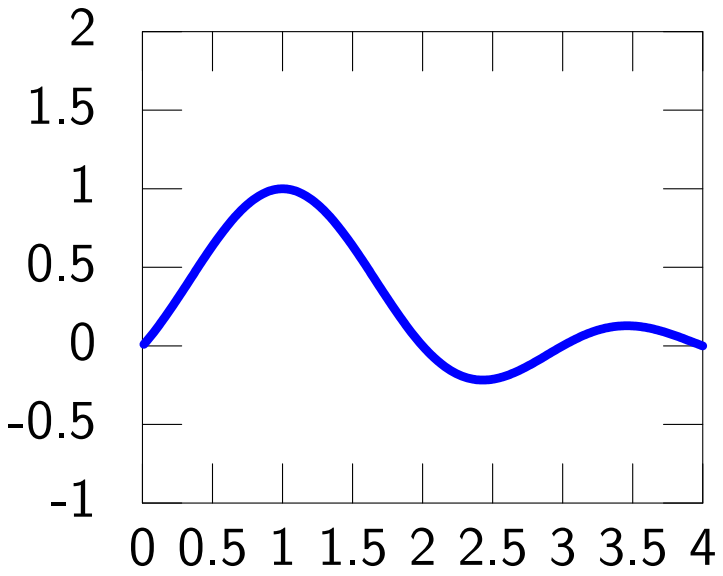
In the remainder, we'll see two settings where decisions are made with either partial information or in a dynamical system. Both of these settings can be formalised with Markov decision processes.

Bandit problems

Bandit problems

Applications

- Efficient optimisation.



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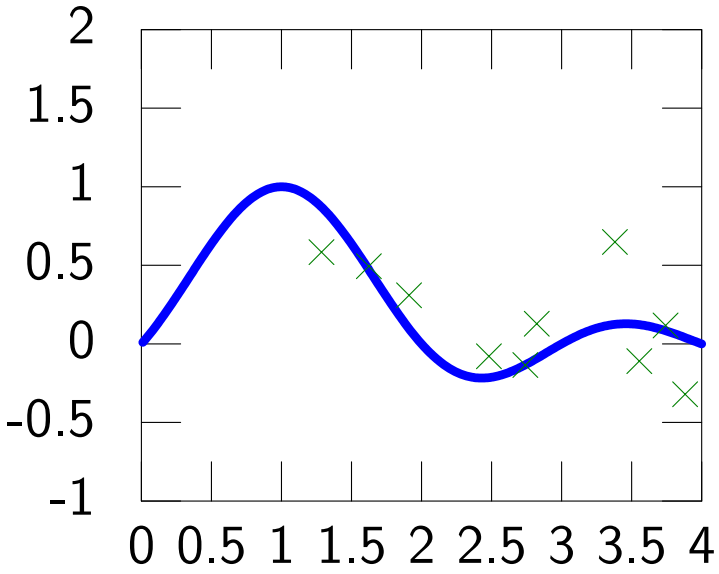

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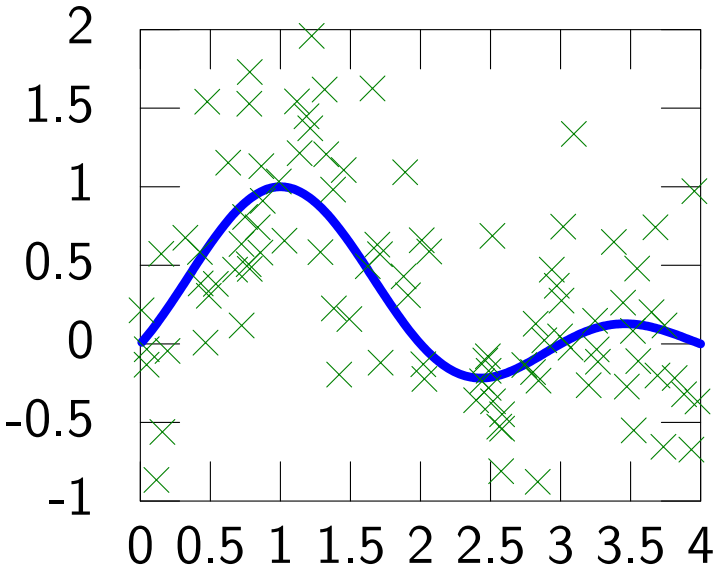

Bandit problems

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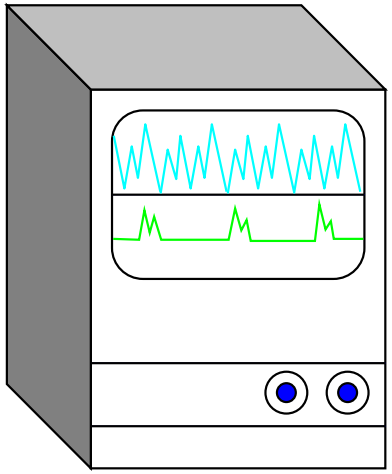

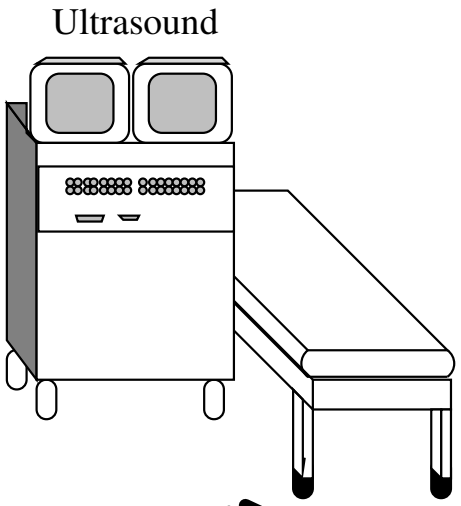

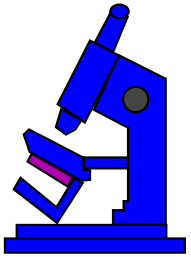

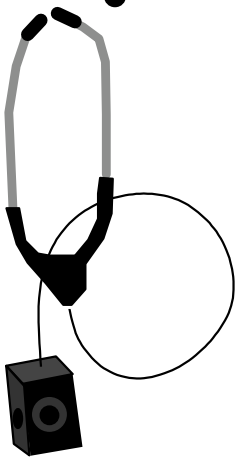

Bandit problems

Applications

- Efficient optimisation.
- Online advertising.
- Clinical trials.
- Robot scientist.

The stochastic n-armed bandit problem

Actions and rewards

- A set of actions $\mathcal{A} = \{1, \ldots, n\}$.
- Each action gives you a random reward with distribution $\mathbb{P}(r_t \mid a_t = i)$.
- The expected reward of the i-th arm is $\rho_i \triangleq \mathbb{E}(r_t \mid a_t = i)$.

Interaction at time t

1. You choose an action $a_t \in \mathcal{A}$.
2. You observe a random reward r_t drawn from the i-th arm.

The utility is the sum of the rewards obtained

$$U \triangleq \sum_t r_t.$$

We must maximise the expected utility, without knowing the values ρ_i.

Policy

Definition 1 (Policies)

A policy π is an algorithm for taking actions given the observed history $h_t \triangleq a_1, r_1, \ldots, a_t, r_t$

$$\mathbb{P}^{\pi}(a_{t+1} \mid h_t)$$

is the probability of the next action a_{t+1}.

Exercise 1

Why should our action depend on the complete history?

A *The next reward depends on all the actions we have taken.*

B *We don't know which arm gives the highest reward.*

C *The next reward depends on all the previous rewards.*

D *The next reward depends on the complete history.*

E *No idea.*

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If $\mathbb{P}^\pi(a_{t+1} \mid \cdot) = 1/n$ for all t, then

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$$\mathbb{E}^\pi U = \mathbb{E}^\pi \left(\sum_{t=1}^{T} r_t \right) = \sum_{t=1}^{T} \mathbb{E}^\pi r_t = \sum_{t=1}^{T} \sum_{i=1}^{n} \frac{1}{n} \rho_i = \frac{T}{n} \sum_{i=1}^{n} \rho_i$$

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$$= \sum_{t=1}^{T} \sum_{a_t \in \mathcal{A}} \mathbb{E}(r_t \mid a_t) \sum_{h_{t-1}} \mathbb{P}^\pi(a_t \mid h_{t-1}) \, \mathbb{P}^\pi(h_{t-1})$$

Contents

Bernoulli bandits

Example 2 (Bernoulli bandits)

Consider n Bernoulli distributions with parameters ω_i $(i = 1, \ldots, n)$ such that $r_t \mid a_t = i \sim \mathcal{Bern}(\omega_i)$. Then,

$$\mathbb{P}(r_t = 1 \mid a_t = i) = \omega_i \qquad \mathbb{P}(r_t = 0 \mid a_t = i) = 1 - \omega_i \tag{1.2}$$

Then the expected reward for the i-th bandit is $\rho_i \triangleq \mathbb{E}(r_t \mid a_t = i) =?$.

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Exercise 1 (The optimal policy under perfect knowledge)

If we know ω_i for all i, what is the best policy?

A *At every step, play the bandit i with the greatest ω_i.*

B *Prefer bandits i with larger ω_i, but play them all.*

C *It depends on the horizon T.*

D *Prefer bandits i which you have played the least so far, but play them all.*

E *It is too complicated.*

The unknown reward case

Say you keep a running average of the reward obtained by each arm

$$\hat{\rho}_{t,i} = R_{t,i}/n_{t,i}$$

where $n_{t,i}$ is the number of times you played arm i and $R_{t,i}$ the total reward received from i so that whenever you play $a_t = i$:

$$R_{t+1,i} = R_{t,i} + r_t, \qquad n_{t+1,i} = n_{t,i} + 1.$$

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Exercise 2 (The optimal policy under imperfect knowledge)

If we just keep travek of the averages $\hat{\rho}_{t,i}$, for all i, what is the best policy?

A *At every step, play the bandit i with the greatest $\hat{\rho}_{t,i}$.*

B *Prefer bandits i with larger $\hat{\rho}_{t,i}$, but play them all.*

C *It depends on the horizon T.*

D *Prefer bandits i with smaller $n_{t,i}$, but play them all.*

E *It is too complicated.*

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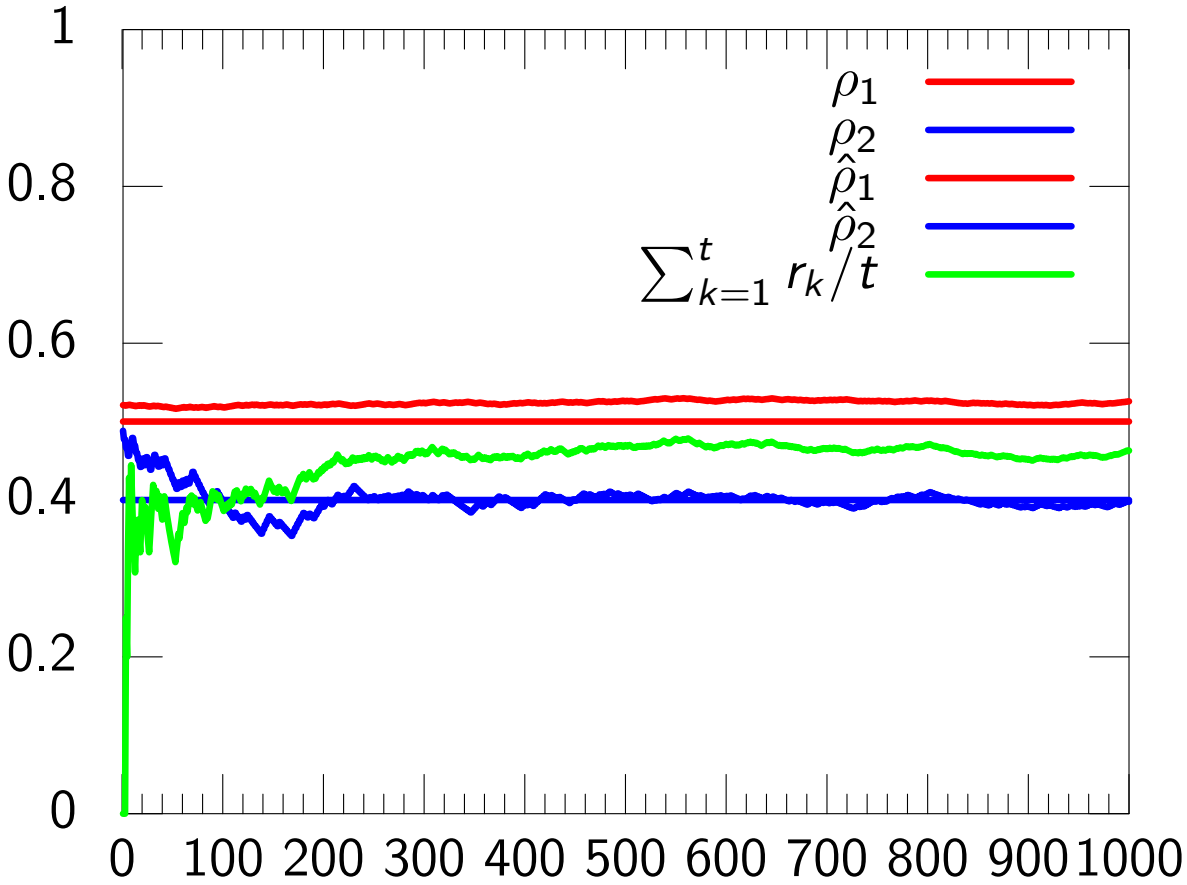
You could choose to play the strategy

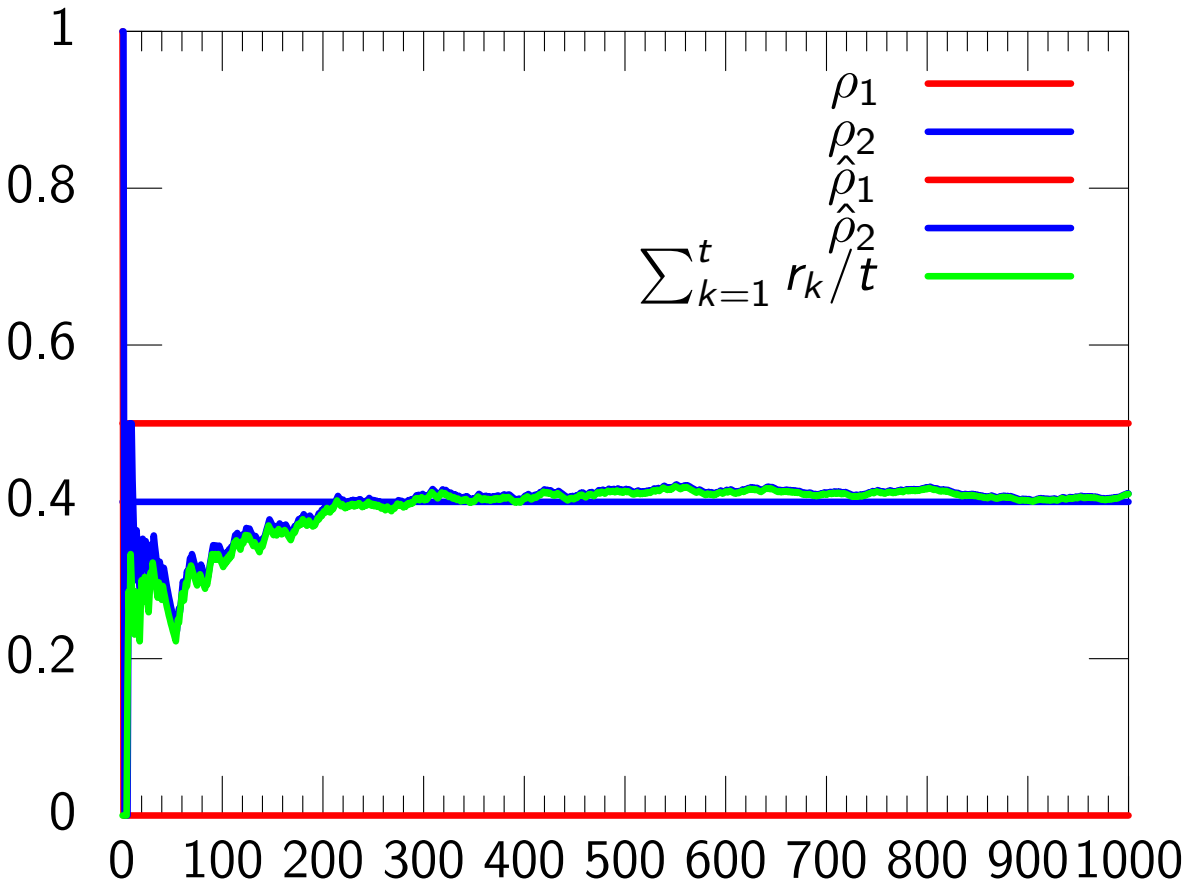
$$a_t = \arg\max_i \hat{\rho}_{t,i}.$$

where we use non-zero initial values $n_{0,i}, R_{0,i}$!

The uniform policy

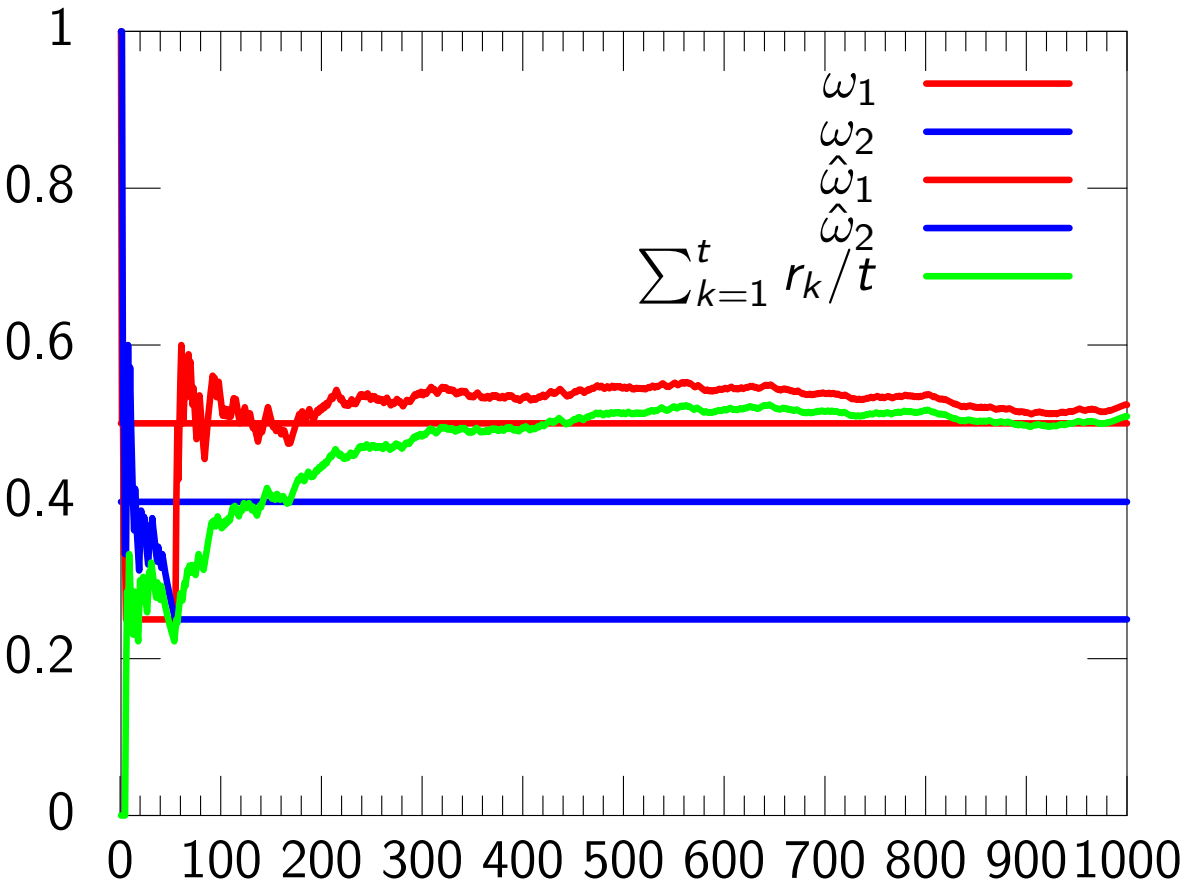


The greedy policy



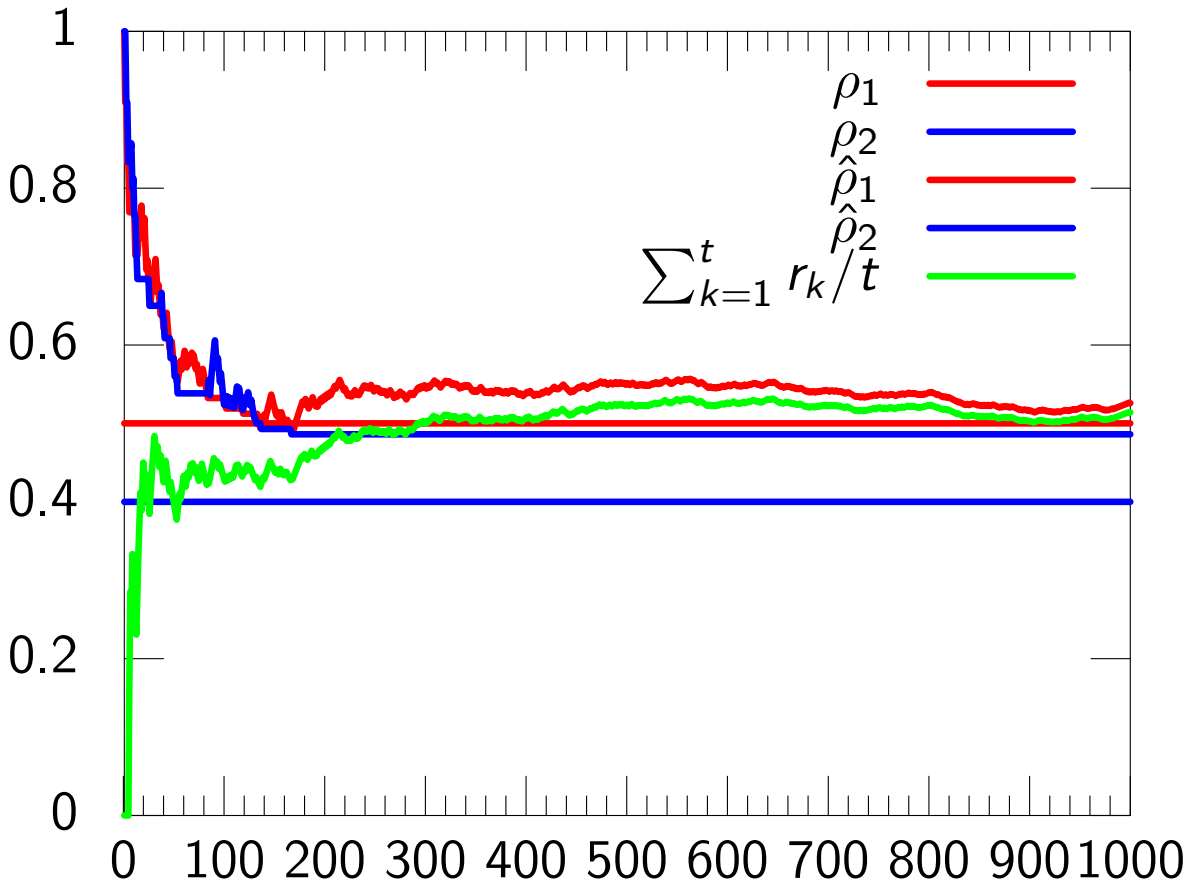
For $n_{0,i} = R_{0,i} = 0$

The greedy policy



For $n_{0,i} = R_{0,i} = 1$

The greedy policy



For $n_{0,i} = R_{0,i} = 10$

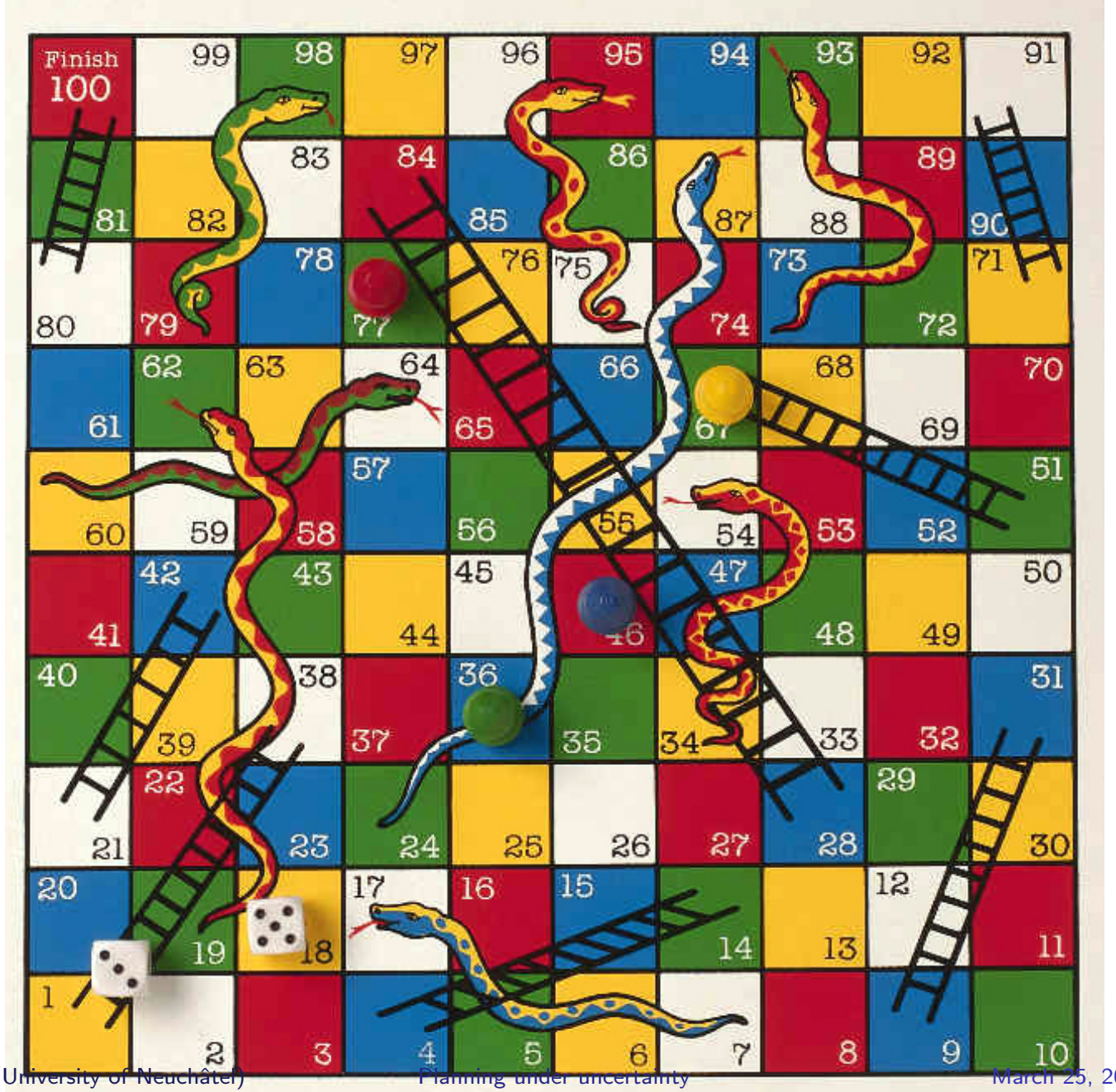
Summary

- Bandit problems are the simplest type of partial information problems.
- Learning policies for such problems must remember the complete history.
- If we know the problem parameters, simple stationary policies are optimal.
- If we don't, then our policies must carefuly balance:
  - Exploration: Learning more about the problem.
  - Exploitation: Using what is already known.

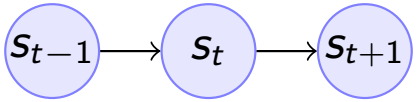
From now on, we focus on the case where the problem is perfectly known.

Contents

A gentle introduction to Markov processes



Markov process



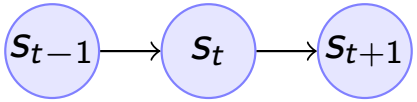
Definition 3 (Markov Process – or Markov Chain)

The sequence $\{s_t \mid t = 1, \ldots\}$ of random variables $s_t : \Omega \to \mathcal{S}$ is a Markov process if

$$\mathbb{P}(s_{t+1} \mid s_t, \ldots, s_1) = \mathbb{P}(s_{t+1} \mid s_t). \tag{2.1}$$

- s_t is state of the Markov process at time t.
- $\mathbb{P}(s_{t+1} \mid s_t)$ is the transition kernel of the process.

Markov process



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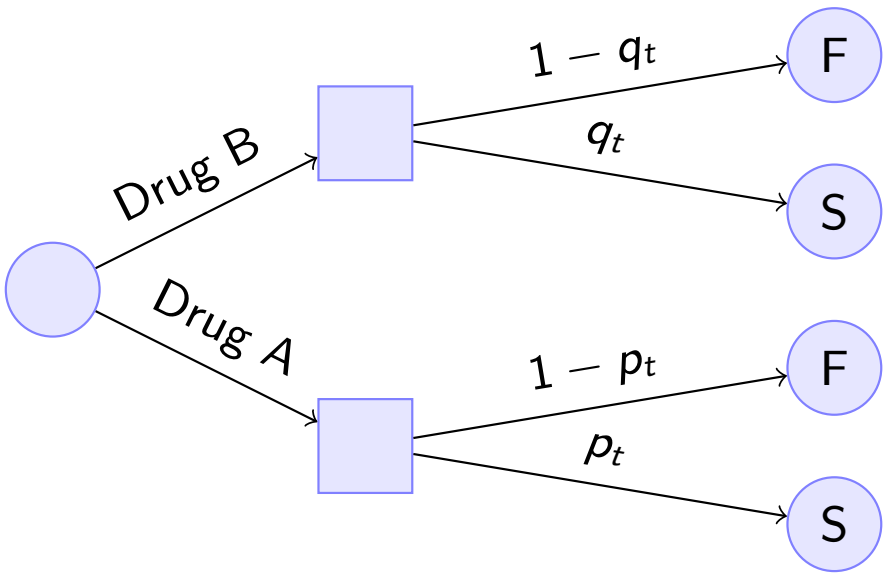
Exercise 2 (Finite state machine with random input)

Let $\omega = \omega_1, \ldots, \omega_t$ be an infinitely long random string of bits, $\Omega = \mathcal{S} = \{0, 1\}$ and:

$$s_{t+1} = s_t \oplus \omega_t.$$

Is s_t a Markov process?

The experiment design problem



s_t a_t r_t s_{t+1}

$$s_t = (\mathbf{R}_t, \mathbf{n}_t)$$

- Probabilities p_t, q_t depend on our information s_t.
- New information (Success, Failure) updates them.
- What drug choices give the most useful information?

Contents

Reinforcement learning

The reinforcement learning problem.

Learning to act in an unknown environment, by interaction and reinforcement.

- The environment has a changing state s_t.
- The agents observes the state s_t.
- The agent takes action a_t.
- It receives rewards r_t.

The goal (informally)

Maximise total reward $\sum_t r_t$

Types of environments

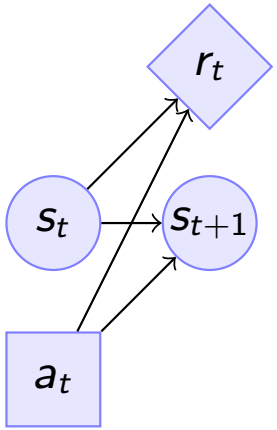
- Markov decision processes (MDPs).
- Partially observable MDPs (POMDPs).
- (Partially observable) Markov games.

Markov decision processes

Markov decision processes (MDP) μ.

At each time step t:

- We observe state $s_t \in \mathcal{S}$.
- We take action $a_t \in \mathcal{A}$.
- We receive a reward $r_t \in \mathbb{R}$.



Markov property of the reward and state distribution

$$\mathbb{P}_\mu(s_{t+1} \mid s_t, a_t) \qquad \text{(Transition distribution)}$$

$$\mathbb{P}_\mu(r_t \mid s_t, a_t) \qquad \text{(Reward distribution)}$$

The agent

The agent's policy π

$$\pi(a_t \mid s_t, \ldots, s_1, a_{t-1}, \ldots, a_1) \qquad \text{(history-dependent policy)}$$

$$\pi(a_t \mid s_t) \qquad \text{(Markov policy)}$$

Definition 4 (Utility)

Given a horizon T, the utility can be defined as

$$U_t \triangleq \sum_{k=0}^{T-t} r_{t+k} \tag{3.1}$$

The agent wants to to find π maximising the expected total future reward

$$\mathbb{E}_\mu^\pi U_t = \mathbb{E}_\mu^\pi \sum_{k=0}^{T-t} r_{t+k}. \qquad \text{(expected utility)}$$

Contents

State value function

$$V^{\pi}_{\mu,t}(s) \triangleq \mathbb{E}^{\pi}_{\mu}(U_t \mid s_t = s) \tag{3.2}$$

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The optimal policy π^*

$$\pi^*(\mu) : V^{\pi^*(\mu)}_{t,\mu}(s) \geq V^{\pi}_{t,\mu}(s), \quad Q^{\pi^*(\mu)}_{t,\mu}(s,a) \geq Q^{\pi}_{t,\mu}(s,a) \qquad \forall \pi, t, s \tag{3.4}$$

dominates all other policies π everywhere in $\mathcal{S}$.

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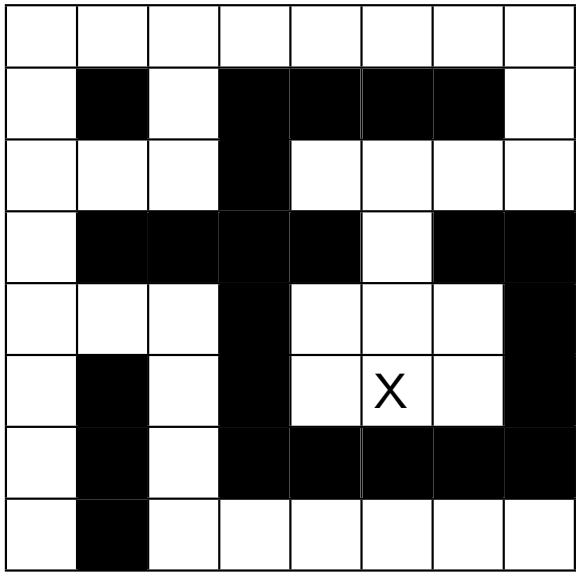
dominates all other policies π everywhere in $\mathcal{S}$. The optimal value function

$$V^*_{t,\mu}(s) \triangleq V^{\pi^*(\mu)}_{t,\mu}(s), \qquad Q^*_{t,\mu}(s,a) \triangleq V^{\pi^*(\mu)}_{t,\mu}(s,a), \tag{3.5}$$

are the value functions of the optimal policy π^*.

Contents

Deterministic shortest-path problems



Properties

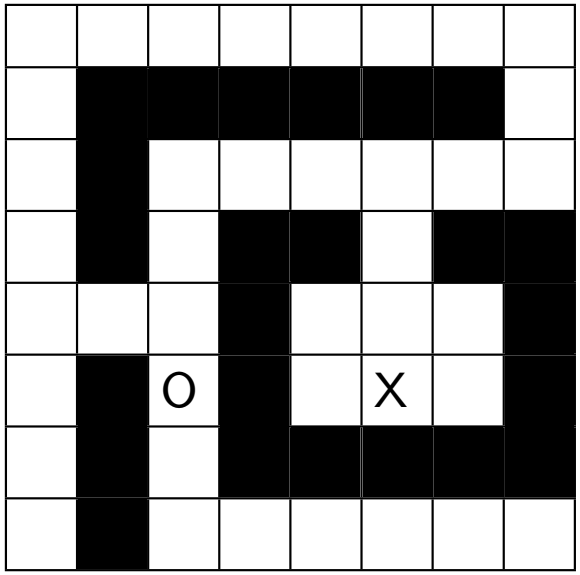
- $T \to \infty$.
- $r_t = -1$ unless $s_t = X$, in which case $r_t = 0$.
- $\mathbb{P}_\mu(s_{t+1} = X | s_t = X) = 1$.
- $\mathcal{A} = \{\mathrm{North}, \mathrm{South}, \mathrm{East}, \mathrm{West}\}$
- Transitions are deterministic and walls block.

14	13	12	11	10	9	8	7
15		13					6
16	15	14		4	3	4	5
17					2		
18	19	20		2	1	2	
19		21		1	0	1	
20		22					
21		23	24	25	26	27	28

Properties

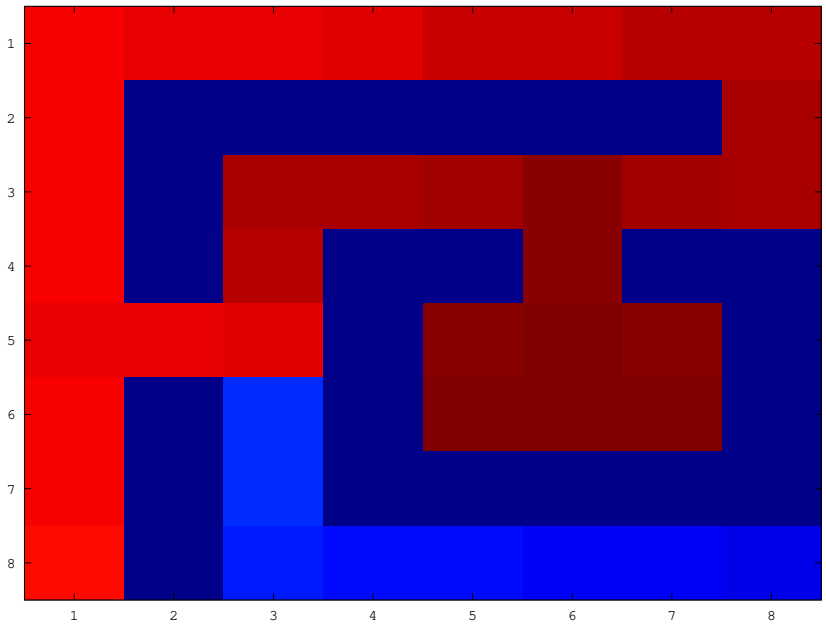
- $\gamma = 1,\ T \to \infty$.
- $r_t = -1$ unless $s_t = X$, in which case $r_t = 0$.
- The length of the shortest path from s equals the negative value of the optimal policy.
- Also called *cost-to-go*.

Stochastic shortest path problem with a pit

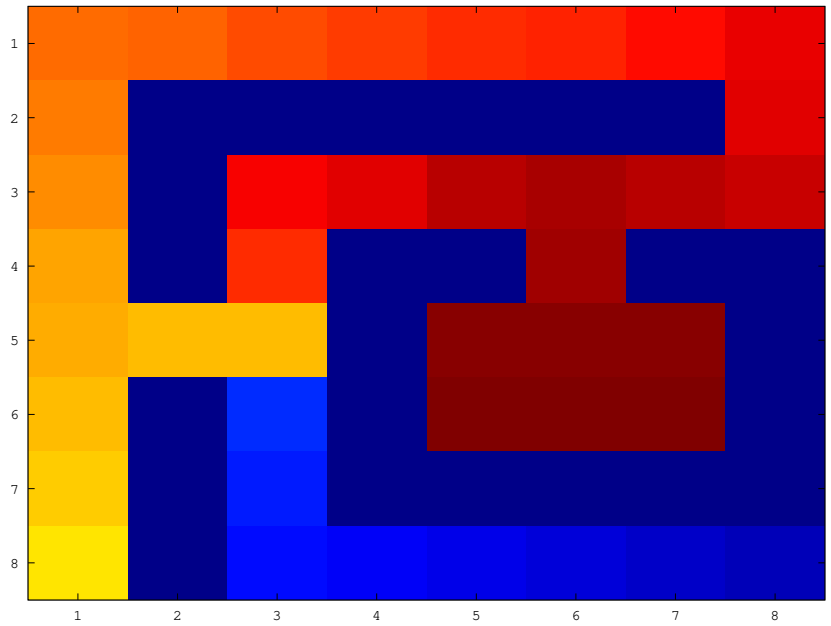


Properties

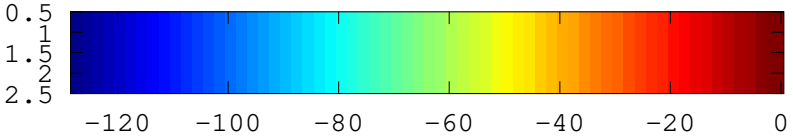
- $T \to \infty$.
- $r_t = -1$, but $r_t = 0$ at X and -100 at O and the problem ends.
- $\mathbb{P}_\mu(s_{t+1} = X | s_t = X) = 1$.
- $\mathcal{A} = \{\text{North}, \text{South}, \text{East}, \text{West}\}$
- Moves to a random direction with probability ω. Walls block.



(a) $\omega = 0.1$



(b) $\omega = 0.5$



(c) value

Exercise 3

- *Why should we only take the shortcut in (a)?*
- *Why does the agent commit suicide at the bottom?*

Contents

How to evaluate a policy

$$V^{\pi}_{\mu,t}(s) \triangleq \mathbb{E}^{\pi}_{\mu}(U_t \mid s_t = s) \tag{4.1}$$

$$\tag{4.2}$$

This derivation directly gives a number of policy evaluation algorithms.

How to evaluate a policy

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How to evaluate a policy

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Monte-Carlo Policy evaluation

for $s \in \mathcal{S}$ **do**

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Monte-Carlo Policy evaluation

for $s \in \mathcal{S}$ **do**
  for $k = 1, \ldots, K$ **do**
    Execute policy π and record total reward K times:

$$\hat{R}_k(s) = \sum_{t=1}^{T} r_{t,k}.$$

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  Calculate estimate:

$$\boldsymbol{v}_t(s) = \frac{1}{K} \sum_{k=1}^{K} \hat{R}_k(s).$$

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Backwards induction policy evaluation

for State $s \in S$, $t = T, \dots, 1$ **do**
Update values of states:

$$v_t(s_t) = \sum_{a_t \in \mathcal{A}} \mathbb{P}^{\pi}(a_t \mid s_t) \left\{ \mathbb{E}_{\mu}(r_t \mid s_t, a_t) + \sum_{s_{t+1} \in \mathcal{S}} \mathbb{P}_{\mu}(s_{t+1} \mid s_t, a_t) v_{t+1}(s_{t+1}) \right\}$$

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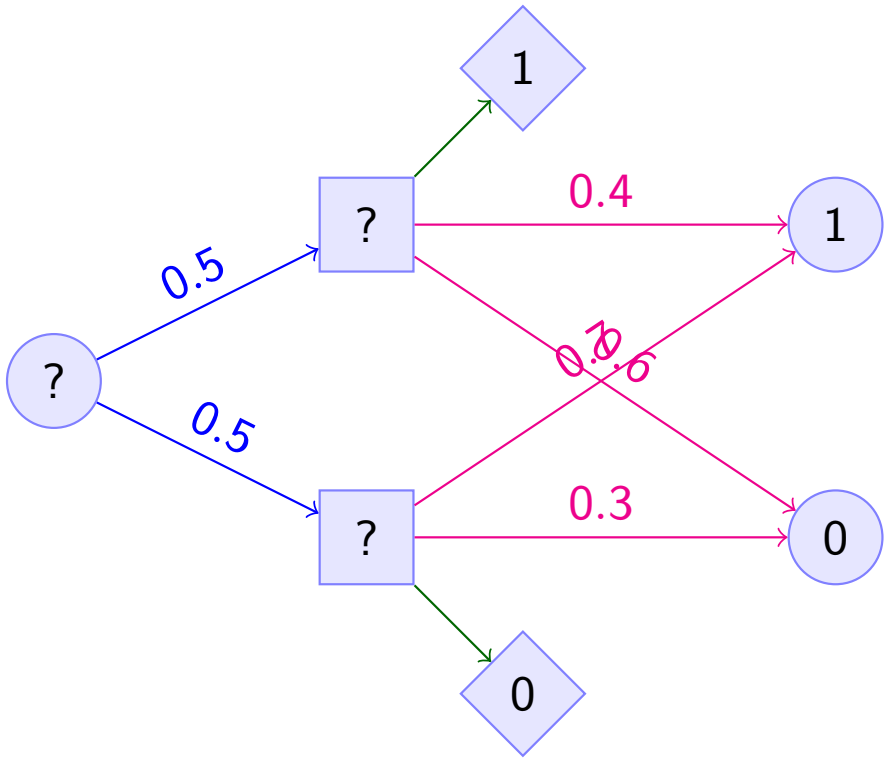

Exercise 4

What is the value $v_t(s_t)$ of the first state?

A *1.4*
B *1.05*
C *1.0*
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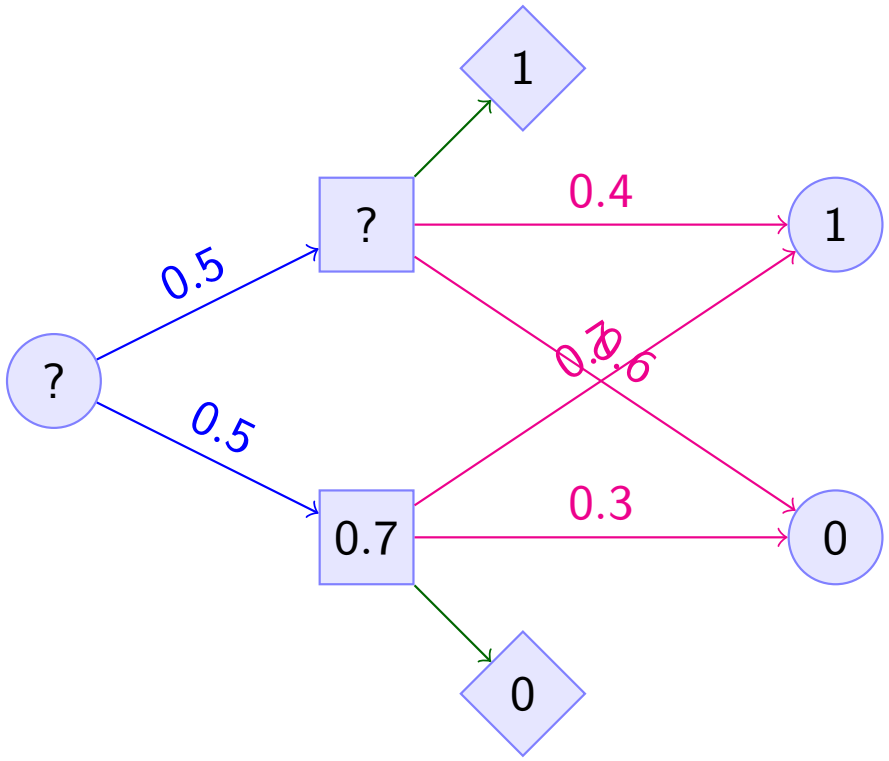

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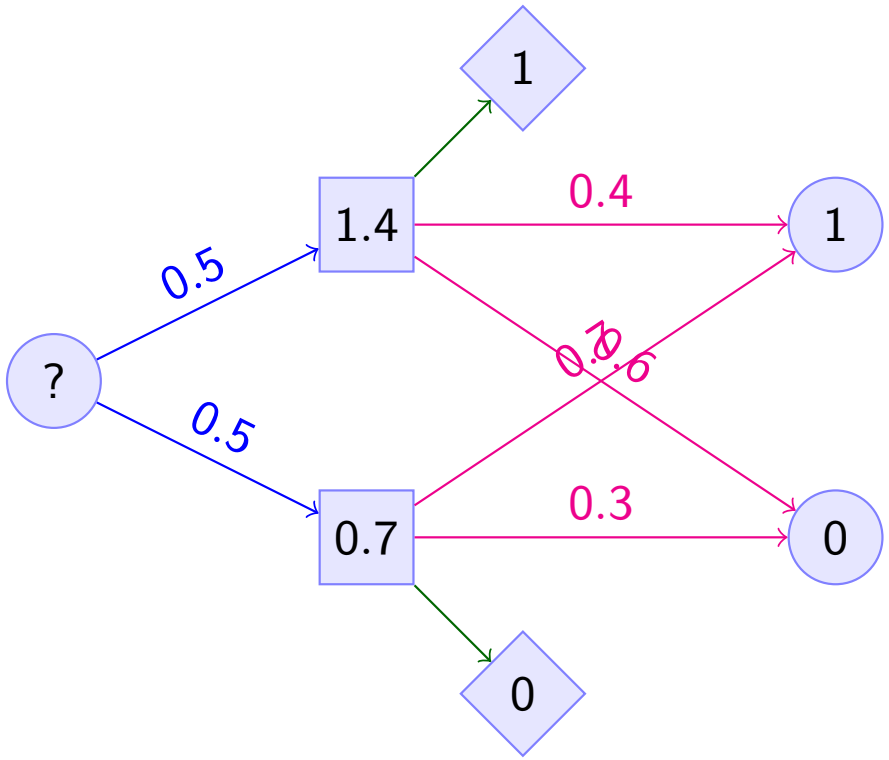

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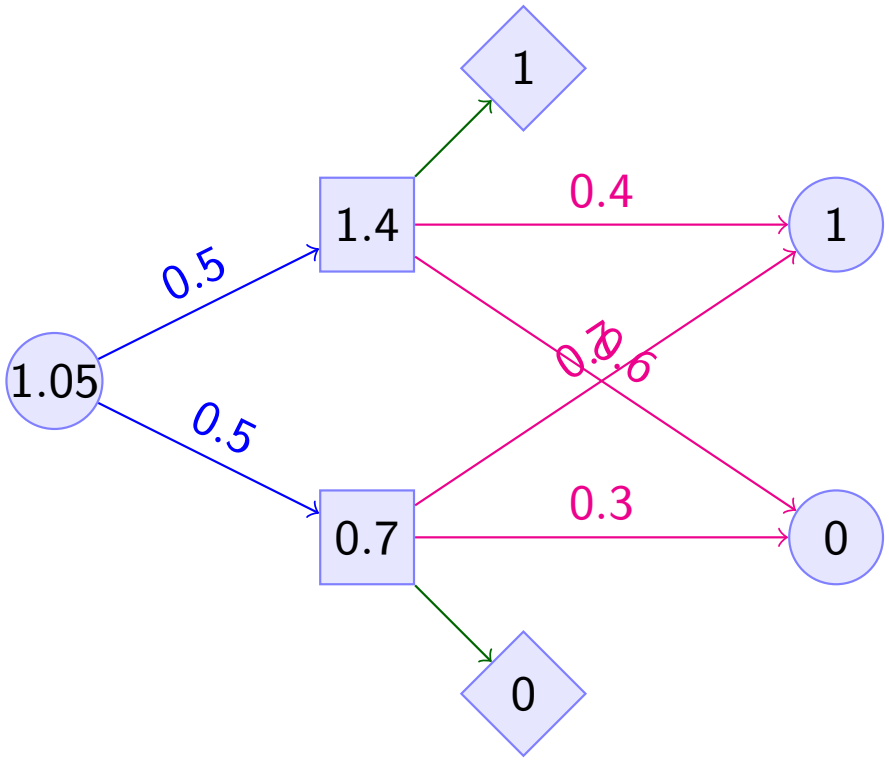

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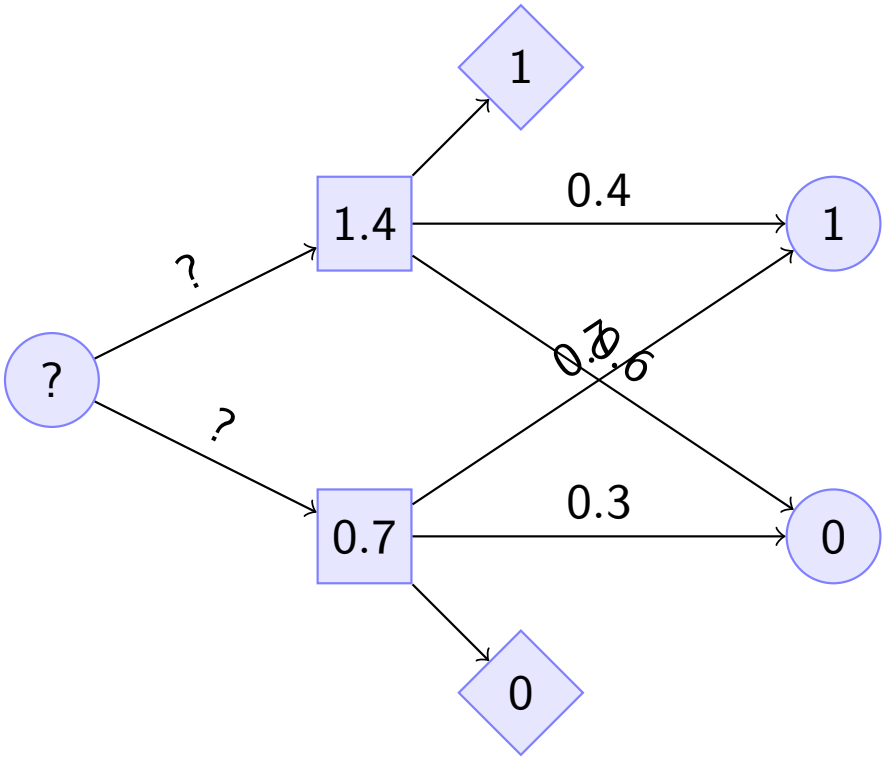

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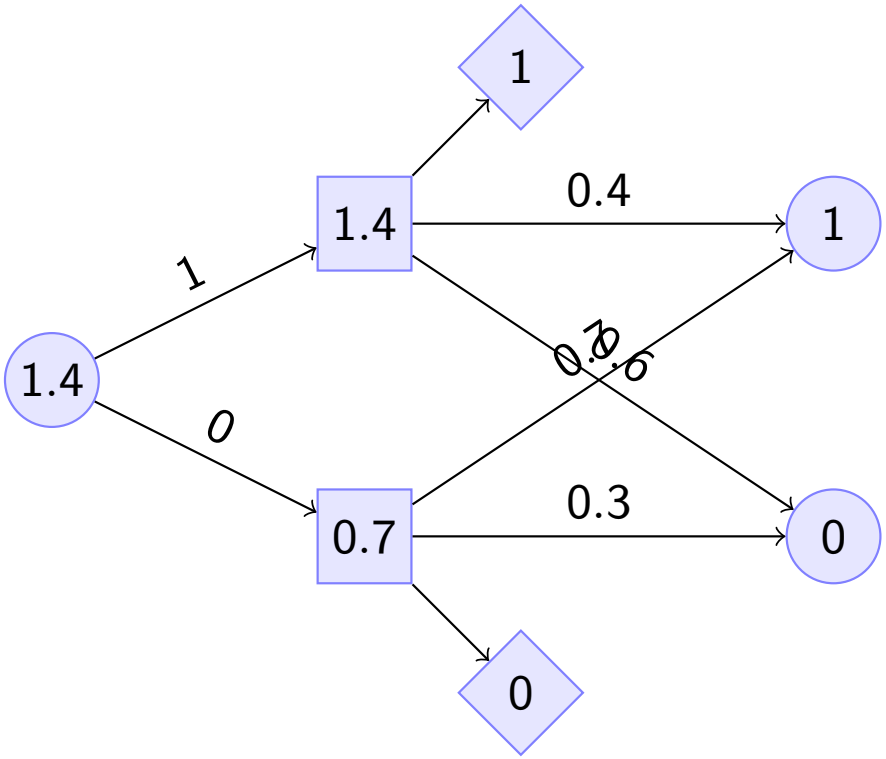

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We will show that BI, returning policy π_{BI} and value $\boldsymbol{v}$, returns the optimal policy and its value function.
We need to prove that $\boldsymbol{v}_t = V_t^*$. First, we will show that $\boldsymbol{v}_t \geq V_t^*$ for all t. We prove this by induction, starting from the base case where $\boldsymbol{v}_T = V_T^*$ trivially. We now need to prove that $\boldsymbol{v}_{t+1} \geq V_{t+1}^* \Rightarrow \boldsymbol{v}_t \geq V_t^*$.

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\end{aligned}$$

Finally, $V^* \geq V^{\pi_{BI}} \geq V^*$.

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Definition 5

A policy π is stationary if $\pi(a_t \mid s_t)$ does not depend on t.

This policy defines the following transition kernel:

$$\boldsymbol{P}_{\mu,\pi}(i,j) = \mathbb{P}^{\pi}_{\mu}(s_{t+1} = j \mid s_t = i) = \sum_a \mathbb{P}^{\pi}_{\mu}(s_{t+1} = j \mid s_t = i, a_t = a)\pi(a_t = a \mid s_t = i)$$

Remark 1

We can use the Markov chain kernel $\boldsymbol{P}_{\mu,\pi}$ to write the expected reward vector as

$$\boldsymbol{v}^{\pi} = \sum_{t=0}^{\infty} \gamma^t \boldsymbol{P}_{\mu,\pi}^t \boldsymbol{r} \tag{5.1}$$

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&= \sum_{t=0}^{\infty} \gamma^t \boldsymbol{P}_{\pi}^t(s) \boldsymbol{r}(s),
\end{aligned}$$

where $\boldsymbol{P}_{\pi}^t(s)$ is the s-th row of the $\boldsymbol{P}_{\pi}^t$ matrix.

Theorem 6

For any stationary policy π, $\boldsymbol{v}^\pi$ is the unique solution of

$$\boldsymbol{v} = \boldsymbol{r} + \gamma \boldsymbol{P}_{\mu,\pi} \boldsymbol{v}. \tag{5.2}$$

In addition, the solution is given by:

$$\boldsymbol{v}^\pi = (\boldsymbol{I} - \gamma \boldsymbol{P}_{\mu,\pi})^{-1} \boldsymbol{r}. \tag{5.3}$$

Matrix norm and spectral radius

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Theorem 7

If $\sigma(\boldsymbol{A}) < 1$, then $(\boldsymbol{I} - \boldsymbol{A})^{-1}$ exists and is given by

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Proof.

Let $\boldsymbol{S}_T = \sum_{t=0}^{T} \boldsymbol{A}^t$. Then $\boldsymbol{A}\boldsymbol{S}_T = \boldsymbol{S}_{T+1} - \boldsymbol{I}$ and:

$$S_T(I - A) = S_T - S_{T+1} + I = I - A^{T+1}$$
$$(I - A)S_T = S_T - S_{T+1} + I = I - A^{T+1}.$$

As $T \to \infty$, $(I - A)S_\infty = S_\infty(I - A) = I$. □

Proof of Theorem ??.

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$$(\boldsymbol{I} - \gamma \boldsymbol{P}_\pi)^{-1} = \lim_{T \to \infty} \sum_{t=0}^{T} (\gamma \boldsymbol{P}_\pi)^t$$

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Proof of Theorem ??.

First note that from (??) one obtains $\boldsymbol{r} = (\boldsymbol{I} - \gamma \boldsymbol{P}_\pi)\boldsymbol{v}$. Since $\|\gamma \boldsymbol{P}_\pi\| = \gamma \cdot \|\boldsymbol{P}_\pi\| = \gamma < 1$, the inverse

$$(\boldsymbol{I} - \gamma \boldsymbol{P}_\pi)^{-1} = \lim_{T \to \infty} \sum_{t=0}^{T} (\gamma \boldsymbol{P}_\pi)^t$$

exists by Theorem ??.

$$\boldsymbol{v} = (\boldsymbol{I} - \gamma \boldsymbol{P}_\pi)^{-1} \boldsymbol{r} = \sum_{t=0}^{\infty} \gamma^t \boldsymbol{P}_\pi^t \boldsymbol{r} = \boldsymbol{v}^\pi.$$

□

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□

The matrix $\boldsymbol{X} = (\boldsymbol{I} - \gamma \boldsymbol{P}_\pi)^{-1}$ is expected number of discounted cumulative visits to each state s, starting from state s' and following policy π. That is,

$$x(s, s') = \mathbb{E}_\mu^\pi \left\{ \sum_{t=0}^{\infty} \gamma^t \, \mathbb{P}_\mu^\pi(s_t = s' \mid s_0 = s) \right\}.$$

Many algorithms rely on an estimation of $\boldsymbol{X}$ for approximating value functions.

Policy evaluation

for $n = 1, 2, \ldots$ and $s \in \mathcal{S}$ **do**
$\boldsymbol{v}_n(s) = \sum_a \pi(a \mid s) \left(r(s,a) + \gamma \sum_{s' \in \mathcal{S}} P_\mu(s' \mid s, a) \boldsymbol{v}_{n-1}(s') \right)$
end for

Policy evaluation

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end for

Value iteration

for $n = 1, 2, \ldots$ and $s \in \mathcal{S}$ **do**
$\boldsymbol{v}_n(s) = \max_a r(s,a) + \gamma \sum_{s' \in \mathcal{S}} P_\mu(s' \mid s, a) \boldsymbol{v}_{n-1}(s')$
end for

Policy evaluation

for $n = 1, 2, \ldots$ and $s \in \mathcal{S}$ **do**
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for $n = 1, 2, \ldots$ and $s \in \mathcal{S}$ **do**
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end for

We can view value iteration as applying the so-called Bellman operator repeatedly.

Definition 8 (Policy and Bellman operator)

Let $\boldsymbol{v} \in \mathcal{V}$, and $\|\boldsymbol{v}\| \triangleq \max \{ |\boldsymbol{v}(s)| \mid s \in \mathcal{S} \}$. We define $\mathscr{L}_\pi : \mathcal{V} \to \mathcal{V}$ and $\mathscr{L} : \mathcal{V} \to \mathcal{V}$ as:

$$\mathscr{L}_\pi \boldsymbol{v} \triangleq \boldsymbol{r}_\pi + \gamma \boldsymbol{P}_\pi \boldsymbol{v}. \qquad \text{(Policy operator)}$$

$$\mathscr{L} \boldsymbol{v} \triangleq \max_\pi \{ \boldsymbol{r}_\pi + \gamma \boldsymbol{P}_\pi \boldsymbol{v} \} = \max_\pi \mathscr{L}_\pi \boldsymbol{v}. \qquad \text{(Bellman operator)}$$

Contents

Bellman optimality condition

Theorem 9

Let $\boldsymbol{v}^* \triangleq \max_\pi \boldsymbol{v}^\pi$ *and* $\boldsymbol{r} \in [0,1]$. *Then for any* $\boldsymbol{v} \in \mathcal{V}$:

(1) *If* $\boldsymbol{v} \geq \mathscr{L}\boldsymbol{v}$, *then* $\boldsymbol{v} \geq \boldsymbol{v}^*$.

(2) *If* $\boldsymbol{v} \leq \mathscr{L}\boldsymbol{v}$, *then* $\boldsymbol{v} \leq \boldsymbol{v}^*$.

(3) *If* $\boldsymbol{v} = \mathscr{L}\boldsymbol{v}$, *then* $\boldsymbol{v} = \max_\pi \boldsymbol{v}^\pi$.

Proof.

We first prove (1). A simple proof by induction over n shows that for any π and any n

$$\boldsymbol{v} \geq \boldsymbol{r} + \gamma \boldsymbol{P}_\pi \boldsymbol{v} \geq \sum_{t=0}^{n-1} \gamma^t \boldsymbol{P}_\pi^t \boldsymbol{r} + \gamma^n \boldsymbol{P}_\pi^n \boldsymbol{v}.$$

□

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We first prove (1). A simple proof by induction over n shows that for any π and any n

$$\boldsymbol{v} \geq \boldsymbol{r} + \gamma \boldsymbol{P}_\pi \boldsymbol{v} \geq \sum_{t=0}^{n-1} \gamma^t \boldsymbol{P}_\pi^t \boldsymbol{r} + \gamma^n \boldsymbol{P}_\pi^n \boldsymbol{v}.$$

Since $\boldsymbol{v}^\pi = \sum_{t=0}^{\infty} \gamma^t \boldsymbol{P}_\pi^t \boldsymbol{r}$, it follows that

$$\boldsymbol{v} - \boldsymbol{v}^\pi \geq \gamma^n \boldsymbol{P}_\pi^n \boldsymbol{v} - \sum_{t=n}^{\infty} \gamma^t \boldsymbol{P}_\pi^t \boldsymbol{r}.$$

□

Bellman optimality condition

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Let $\boldsymbol{v}^* \triangleq \max_\pi \boldsymbol{v}^\pi$ *and* $\boldsymbol{r} \in [0,1]$. *Then for any* $\boldsymbol{v} \in \mathcal{V}$:

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Proof.

Since $\boldsymbol{v}^\pi = \sum_{t=0}^\infty \gamma^t \boldsymbol{P}_\pi^t \boldsymbol{r}$, it follows that

$$\boldsymbol{v} - \boldsymbol{v}^\pi \geq \gamma^n \boldsymbol{P}_\pi^n \boldsymbol{v} - \sum_{t=n}^\infty \gamma^t \boldsymbol{P}_\pi^t \boldsymbol{r}.$$

Let $\boldsymbol{e}$ be the vector with $\boldsymbol{e}(s) = 1$ for all s. Then, as rewards are assumed to be in $[0,1]$,

$$\sum_{k=n}^\infty \gamma^k \boldsymbol{P}_\pi^k \boldsymbol{r} \leq \frac{\gamma^n \boldsymbol{e}}{1-\gamma},$$

which for $n \to \infty$ approaches $\mathbf{0}$. It follows that for any π

$$\boldsymbol{v} \geq \boldsymbol{v}^\pi,$$

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which for $n \to \infty$ approaches $\mathbf{0}$. It follows that for any π

$$\boldsymbol{v} \geq \boldsymbol{v}^\pi,$$

and hence also $\boldsymbol{v} \geq \boldsymbol{v}^*$, which completes the proof of (1). A similar argument shows (2), which together with (1) then implies (3). □

Convergence of value iteration

Theorem 10

We want to show that repeated application of the Bellman operator always converges to the optimal value, independent of the initial value $\boldsymbol{v}_0$.

1. *There is a unique $\boldsymbol{v}^* \in \mathcal{V}$ such that $\mathscr{L}\boldsymbol{v}^* = \boldsymbol{v}^*$ and $\boldsymbol{v}^* = \max_\pi \boldsymbol{v}^\pi$.*
2. *For any stationary policy π, there is a unique $\boldsymbol{v} \in \mathcal{V}$ such that $\mathscr{L}_\pi \boldsymbol{v} = \boldsymbol{v}$ and $\boldsymbol{v} = \boldsymbol{v}^\pi$.*

Proof.

We first prove that $\mathscr{L}$ is a contraction. The Banach Fixed Point Theorem then tells us there is a unique $\boldsymbol{v}^* \in \mathcal{V}$ such that $\mathscr{L}\boldsymbol{v}^* = \boldsymbol{v}^*$. This is also the optimal value function due to Theorem **??**.
The second part of the theorem follows from the first part when considering only a single policy π (which then is optimal). □

Contraction mapping

Definition 11

For a Banach space $\mathcal{X}$ (i.e., a complete, normed linear space) we say that $T : \mathcal{X} \to \mathcal{X}$ is a *contraction mapping*, if there is $\gamma \in [0, 1)$ such that $\|T\boldsymbol{u} - T\boldsymbol{v}\| \leq \gamma\|\boldsymbol{u} - \boldsymbol{v}\|$ for all $\boldsymbol{u}, \boldsymbol{v} \in \mathcal{X}$.

Theorem 12 (Banach fixed-point theorem)

Given a Banach space $\mathcal{X}$ and a contraction mapping $T : \mathcal{X} \to \mathcal{X}$, it holds that:

1. *There is a unique $\boldsymbol{u}^* \in \mathcal{X}$ with $T\boldsymbol{u}^* = \boldsymbol{u}^*$.*
2. *For any $\boldsymbol{u}^0 \in \mathcal{X}$ the sequence $(\boldsymbol{u}_n)_{n\geq 0}$ defined by $\boldsymbol{u}_{n+1} = T\boldsymbol{u}_n = T^{n+1}\boldsymbol{u}_0$ converges to $\boldsymbol{u}^*$.*

Proof.

First note that for any $m \geq 1$

$$\|\boldsymbol{u}_{n+m} - \boldsymbol{u}_n\| \leq \sum_{k=0}^{m-1} \|\boldsymbol{u}_{n+k+1} - \boldsymbol{u}_{n+k}\| = \sum_{k=0}^{m-1} \|T^{n+k}\boldsymbol{u}_1 - T^{n+k}\boldsymbol{u}_0\|$$

□

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First note that for any $m \geq 1$

$$\begin{aligned}\|\boldsymbol{u}_{n+m} - \boldsymbol{u}_n\| &\leq \sum_{k=0}^{m-1} \|\boldsymbol{u}_{n+k+1} - \boldsymbol{u}_{n+k}\| = \sum_{k=0}^{m-1} \|T^{n+k}\boldsymbol{u}_1 - T^{n+k}\boldsymbol{u}_0\| \\ &\leq \sum_{k=0}^{m-1} \gamma^{n+k}\|\boldsymbol{u}_1 - \boldsymbol{u}_0\| = \frac{\gamma^n(1-\gamma^m)}{1-\gamma}\|\boldsymbol{u}_1 - \boldsymbol{u}_0\|.\end{aligned}$$

□

Theorem 12 (Banach fixed-point theorem)

Given a Banach space $\mathcal{X}$ and a contraction mapping $T : \mathcal{X} \to \mathcal{X}$, it holds that:

1. *There is a unique $\boldsymbol{u}^* \in \mathcal{X}$ with $T\boldsymbol{u}^* = \boldsymbol{u}^*$.*
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First note that for any $m \geq 1$

$$\begin{aligned}\|\boldsymbol{u}_{n+m} - \boldsymbol{u}_n\| &\leq \sum_{k=0}^{m-1} \|\boldsymbol{u}_{n+k+1} - \boldsymbol{u}_{n+k}\| = \sum_{k=0}^{m-1} \|T^{n+k}\boldsymbol{u}_1 - T^{n+k}\boldsymbol{u}_0\| \\ &\leq \sum_{k=0}^{m-1} \gamma^{n+k}\|\boldsymbol{u}_1 - \boldsymbol{u}_0\| = \frac{\gamma^n(1-\gamma^m)}{1-\gamma}\|\boldsymbol{u}_1 - \boldsymbol{u}_0\|.\end{aligned}$$

So $\forall \epsilon > 0$, $\exists n$ s.t $\|\boldsymbol{u}^{n+m} - \boldsymbol{u}^n\| \leq \epsilon$. Since $\mathcal{X}$ is a Banach space, the sequence has a limit $\boldsymbol{u}^* \in \mathcal{X}$. □

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Proof.

Next, we show that this $\boldsymbol{u}^*$ is indeed a fixed point of T. Indeed, we have

$$\|T\boldsymbol{u}^* - \boldsymbol{u}^*\| \leq \|T\boldsymbol{u}^* - \boldsymbol{u}_n\| + \|\boldsymbol{u}_n - \boldsymbol{u}^*\| \tag{6.1}$$

□

Theorem 12 (Banach fixed-point theorem)

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As $\|T\boldsymbol{u}^* - \boldsymbol{u}_n\| = \|T\boldsymbol{u}^* - T\boldsymbol{u}_{n-1}\| \leq \gamma\|\boldsymbol{u}^* - \boldsymbol{u}_{n-1}\|$, both terms on the rhs approach 0 for $n \to \infty$, whence we have $\|T\boldsymbol{u}^* - \boldsymbol{u}^*\| = 0$ and $T\boldsymbol{u}^* = \boldsymbol{u}^*$. □

Theorem 12 (Banach fixed-point theorem)

Given a Banach space $\mathcal{X}$ and a contraction mapping $T : \mathcal{X} \to \mathcal{X}$, it holds that:

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Proof.

We conclude by showing uniqueness. If $\boldsymbol{u}', \boldsymbol{u}''$ are both fixed points then by the contraction property

$$\|\boldsymbol{u}' - \boldsymbol{u}''\| = \|T\boldsymbol{u}' - T\boldsymbol{u}''\| \leq \gamma\|\boldsymbol{u}' - \boldsymbol{u}''\|,$$

whence it follows that $\|\boldsymbol{u}' - \boldsymbol{u}''\| = 0$ and $\boldsymbol{u}' = \boldsymbol{u}''$. □

Theorem 13

For $\gamma \in [0,1)$ the Bellman operator $\mathscr{L}$ is a contraction mapping in $\mathcal{V}$.

Proof.

Let $\boldsymbol{v}, \boldsymbol{v}' \in \mathcal{V}$. Consider $s \in \mathcal{S}$ such that $\mathscr{L}\boldsymbol{v}(s) \geq \mathscr{L}\boldsymbol{v}'(s)$, and let

$$a_s^* \in \arg\max_{a \in \mathcal{A}} \left\{ r(s) + \sum_{s' \in \mathcal{S}} \gamma \, p(s' \mid s, a) \, \boldsymbol{v}(s') \right\}.$$

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Then, as a_s^* is optimal for $\boldsymbol{v}(s)$, but not necessarily for $\boldsymbol{v}'(s)$, we have

$$0 \leq \mathscr{L}\boldsymbol{v}(s) - \mathscr{L}\boldsymbol{v}'(s) \leq \gamma \|\boldsymbol{v} - \boldsymbol{v}'\|.$$

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Repeating the argument for s such that $\mathscr{L}\boldsymbol{v}(s) \leq \mathscr{L}\boldsymbol{v}'(s)$, we obtain

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$$|\mathscr{L}\boldsymbol{v}(s) - \mathscr{L}\boldsymbol{v}'(s)| \leq \gamma \|\boldsymbol{v} - \boldsymbol{v}'\|.$$

It follows that $\|\mathscr{L}\boldsymbol{v} - \mathscr{L}\boldsymbol{v}'\| = \max_s |\mathscr{L}\boldsymbol{v}(s) - \mathscr{L}\boldsymbol{v}'(s)| \leq \gamma \|\boldsymbol{v} - \boldsymbol{v}'\|$. □

We note that is easy to adapt this proof to show that $\mathscr{L}_\pi$ is a contraction, too.

Theorem 14

Value iteration satisfies:

1. $\lim_{n\to\infty} \|\boldsymbol{v}_n - \boldsymbol{v}^*\| = 0$.
2. *For each* $\epsilon > 0$ *there exists* $N_\epsilon < \infty$ *such that for all* $n \geq N_\epsilon$*:*
   - 2.1 $\|\boldsymbol{v}_{n+1} - \boldsymbol{v}_n\| \leq \epsilon(1-\gamma)/2\gamma$.
   - 2.2 $\|\boldsymbol{v}_n - \boldsymbol{v}^*\| < \epsilon/2$.
   - 2.3 *The policy* π_n *is* ϵ*-optimal, i.e.,* $\boldsymbol{v}^{\pi_n}(s) \geq \boldsymbol{v}^*(s) - \epsilon$ *for all states* s.

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Proof.

Statements 1 and 2(a) follow from Theorems **??** and **??** from the previous section. Now note that

$$\|\boldsymbol{v}^{\pi_n} - \boldsymbol{v}^*\| \leq \|\boldsymbol{v}^{\pi_n} - \boldsymbol{v}_{n+1}\| + \|\boldsymbol{v}_{n+1} - \boldsymbol{v}^*\|.$$

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Value iteration satisfies:

1. $\lim_{n\to\infty} \|\boldsymbol{v}_n - \boldsymbol{v}^*\| = 0$.
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Proof.

Statements 1 and 2(a) follow from Theorems **??** and **??** from the previous section. Now note that

$$\|\boldsymbol{v}^{\pi_n} - \boldsymbol{v}^*\| \leq \|\boldsymbol{v}^{\pi_n} - \boldsymbol{v}_{n+1}\| + \|\boldsymbol{v}_{n+1} - \boldsymbol{v}^*\|.$$

For the first term we have by Theorems **??** and **??**

$$\|\boldsymbol{v}^{\pi_n} - \boldsymbol{v}_{n+1}\| \leq \frac{\gamma}{1-\gamma}\|\boldsymbol{v}_{n+1} - \boldsymbol{v}_n\|, \qquad \|\boldsymbol{v}_{n+1} - \boldsymbol{v}^*\| \leq \frac{\gamma}{1-\gamma}\|\boldsymbol{v}_{n+1} - \boldsymbol{v}_n\|,$$

and 2(b) as well as 2(c) follow from 2(a). □

Theorem 15

Let $\boldsymbol{v}_0 = \mathbf{0}$ and assume that rewards are bounded in $[0, 1]$. Then

$$\|\boldsymbol{v}_n - \boldsymbol{v}^*\| \leq \frac{\gamma^n}{1-\gamma}, \text{ and } \quad \|\boldsymbol{v}^{\pi_n} - \boldsymbol{v}^*\| \leq \frac{2\gamma^n}{1-\gamma}.$$

Proof.

For the first part, note that

$$\|\boldsymbol{v}_0 - \boldsymbol{v}^*\| = \|\boldsymbol{v}^*\| \leq \frac{1}{1-\gamma} = \frac{\gamma^0}{1-\gamma}.$$

Proceeding by induction over n proves the first claim, as by the contraction property of Theorem **??** we have

$$\|\boldsymbol{v}_{n+1} - \boldsymbol{v}^*\| = \|\mathscr{L}\boldsymbol{v}_n - \mathscr{L}\boldsymbol{v}^*\| \leq \gamma\|\boldsymbol{v}_n - \boldsymbol{v}^*\|.$$

The second part can be shown similarly to 2(c) of Theorem **??**. □

Contents

Policy Iteration

input μ, $\mathcal{S}$.
Initialise $\boldsymbol{v}_0$.
for $n = 1, 2, \ldots$ **do**
  $\pi_n = \arg\max_\pi \{\boldsymbol{r} + \gamma \boldsymbol{P}_\pi \boldsymbol{v}_{n-1}\}$ `// policy improvement`
  $\boldsymbol{v}_n = V_\mu^{\pi_n}$ `// policy evaluation`
  break if $\pi_n = \pi_{n-1}$.
end for
Return $\pi_n, \boldsymbol{v}_n$.

Value Iteration re-written

input μ, $\mathcal{S}$.
Initialise $\boldsymbol{v}_0$.
for $n = 1, 2, \ldots$ **do**
  $\pi_n = \arg\max_\pi \{\boldsymbol{r} + \gamma \boldsymbol{P}_\pi \boldsymbol{v}_{n-1}\}$ `// policy improvement`
  $\boldsymbol{v}_n = \{\boldsymbol{r} + \gamma \boldsymbol{P}_{\pi_n} \boldsymbol{v}_{n-1}\}$ `// one-step policy evaluation`
end for
Return $\pi_n, \boldsymbol{v}_n$.

Between value iteration and policy iteration

input μ, parameter m
Initialization: Choose some $\boldsymbol{v}_0 \in \mathcal{V}$.
for $n = 1, 2, \ldots, N$ **do**
  $\pi_n = \arg\max_\pi \{\boldsymbol{r} + \gamma \boldsymbol{P}_\pi \boldsymbol{v}_{n-1}\}$ `// policy improvement`
  $\boldsymbol{v}_n = \mathscr{L}_{\pi_n^m} \boldsymbol{v}_{n-1}$ `// partial policy evaluation`
end for
return $\pi_N, \boldsymbol{v}_N$

Theorem 16

For value vectors $\boldsymbol{v}_n, \boldsymbol{v}_{n+1}$ generated by policy iteration it holds that $\boldsymbol{v}_n \leq \boldsymbol{v}_{n+1}$.

Proof.

The one-step policy improvement gives:

□

Theorem 16

For value vectors $\boldsymbol{v}_n, \boldsymbol{v}_{n+1}$ generated by policy iteration it holds that $\boldsymbol{v}_n \leq \boldsymbol{v}_{n+1}$.

Proof.

The one-step policy improvement gives:

$$\boldsymbol{r} + \gamma \boldsymbol{P}_{\pi_{n+1}} \boldsymbol{v}_n$$

□

Theorem 16

For value vectors $\boldsymbol{v}_n, \boldsymbol{v}_{n+1}$ generated by policy iteration it holds that $\boldsymbol{v}_n \leq \boldsymbol{v}_{n+1}$.

Proof.

The one-step policy improvement gives:

$$\boldsymbol{r} + \gamma \boldsymbol{P}_{\pi_{n+1}} \boldsymbol{v}_n \geq \boldsymbol{r} + \gamma \boldsymbol{P}_{\pi_n} \boldsymbol{v}_n$$

□

Theorem 16

For value vectors $\boldsymbol{v}_n, \boldsymbol{v}_{n+1}$ generated by policy iteration it holds that $\boldsymbol{v}_n \leq \boldsymbol{v}_{n+1}$.

Proof.

The one-step policy improvement gives:

$$\boldsymbol{r} + \gamma \boldsymbol{P}_{\pi_{n+1}} \boldsymbol{v}_n \geq \boldsymbol{r} + \gamma \boldsymbol{P}_{\pi_n} \boldsymbol{v}_n = \boldsymbol{v}_n$$

□

Theorem 16

For value vectors $\boldsymbol{v}_n, \boldsymbol{v}_{n+1}$ generated by policy iteration it holds that $\boldsymbol{v}_n \leq \boldsymbol{v}_{n+1}$.

Proof.

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Rearranging, we get $\boldsymbol{r} \geq (\boldsymbol{I} - \gamma \boldsymbol{P}_{\pi_{n+1}}) \boldsymbol{v}_n$

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Rearranging, we get $\boldsymbol{r} \geq (\boldsymbol{I} - \gamma \boldsymbol{P}_{\pi_{n+1}})\, \boldsymbol{v}_n$ and hence

$$(\boldsymbol{I} - \gamma \boldsymbol{P}_{\pi_{n+1}})^{-1} \boldsymbol{r} \;\geq\; \boldsymbol{v}_n.$$

□

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For value vectors $\boldsymbol{v}_n, \boldsymbol{v}_{n+1}$ generated by policy iteration it holds that $\boldsymbol{v}_n \leq \boldsymbol{v}_{n+1}$.

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$$(\boldsymbol{I} - \gamma \boldsymbol{P}_{\pi_{n+1}})^{-1} \boldsymbol{r} \;\geq\; \boldsymbol{v}_n.$$

The left hand side equals $\boldsymbol{v}_{n+1}$, which completes the proof. □

Corollary 17

Policy iteration terminates after a finite number of iterations and returns an optimal policy.

Proof.

- There is only a finite number of policies.

□

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- There is only a finite number of policies.
- Policy iteration is monotonically improving.

□

Corollary 17

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Proof.

- There is only a finite number of policies.
- Policy iteration is monotonically improving.
- Finally, the last iteration satisfies the Bellman optimality equation,

$$\boldsymbol{v}_n = \max_{\pi} \{\boldsymbol{r} + \gamma \boldsymbol{P}_\pi \boldsymbol{v}_n\}.$$

□

Temporal differences

Definition 18

The *difference operator* is defined as

$$\mathscr{B}\boldsymbol{v} \triangleq \max_{\pi} \{\boldsymbol{r} + (\gamma \boldsymbol{P}_{\pi} - \boldsymbol{I})\boldsymbol{v}\} = \mathscr{L}\boldsymbol{v} - \boldsymbol{v}.$$

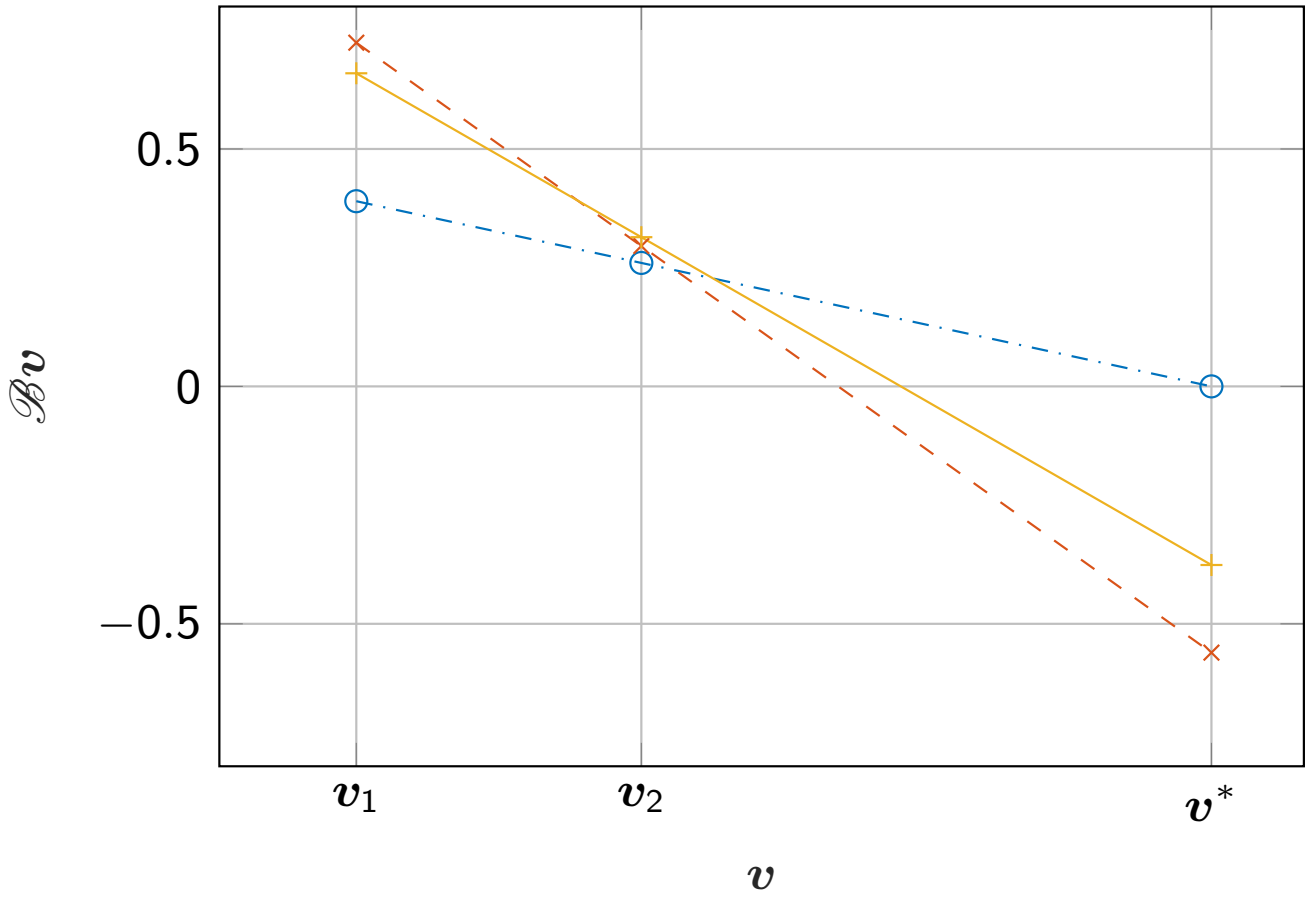
Accordingly, the Bellman optimality equation can be rewritten as

$$\mathscr{B}\boldsymbol{v} = \boldsymbol{0}.$$

Since the identity matrix plays no role in the maximisation, we can rewrite policy iteration

$$\pi_n = \arg\max_{\pi} \{\boldsymbol{r} + (\gamma \boldsymbol{P}_{\pi} - \boldsymbol{I})\boldsymbol{v}_{n-1}\}$$

$$\boldsymbol{v}_n = \boldsymbol{v}_{n-1} - (\gamma \boldsymbol{P}_{\pi} - \boldsymbol{I})^{-1} \mathscr{B}\boldsymbol{v}_{n-1}$$



The improvement of greedy policies π^*, π_1, π_2 on value functions $\boldsymbol{v}^*, \boldsymbol{v}_1, \boldsymbol{v}_2$.

Temporal difference policy iteration

In order to update the value from $\boldsymbol{v}_n$ to $\boldsymbol{v}_{n+1}$ we rely on the temporal difference error:

$$d_n(s, s') = \big(\boldsymbol{r}(s) + \gamma \boldsymbol{v}_n(s')\big) - \boldsymbol{v}_n(s).$$

$$\boldsymbol{v}_{n+1} = \boldsymbol{v}_n + \boldsymbol{\tau}_n, \text{ where}$$

$$\boldsymbol{\tau}_n(s) = \sum_{t=0}^{\infty} \mathbb{E}_{\pi_n}\big[(\gamma\lambda)^t d_n(s_t, s_{t+1}) \mid s_0 = s\big].$$

Temporal difference Policy Iteration

input μ, parameter $\lambda \in [0, 1]$
Initialization: Choose some $\boldsymbol{v}_0 \in \mathcal{V}$. Set $n = 0$.
for $n = 1, \ldots$ **do**
  $\pi_n = \arg\max_\pi \{\boldsymbol{r} + \gamma \boldsymbol{P}_\pi \boldsymbol{v}_{n-1}\}$ `// policy improvement`
  $\boldsymbol{v}_n = \boldsymbol{v}_{n-1} + \boldsymbol{\tau}_{n-1}$ `// temporal difference update`
  break if $\pi_n = \pi_{n-1}$
end for
return $\pi_n, \boldsymbol{v}_n$

Linear programming

Primal: Minimum upper bound on the value

For some state distribution $\boldsymbol{y} \in \mathbb{\Delta}(\mathcal{S})$,

$$\min_{\boldsymbol{v}} \boldsymbol{y}^\top \boldsymbol{v},$$

such that for all $s \in \mathcal{S}, a \in \mathcal{A}$

$$\boldsymbol{v}(s) - \gamma \boldsymbol{p}_{s,a}^\top \boldsymbol{v} \geq r(s).$$

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Dual: State visitation

$$\max_{\boldsymbol{x}} \sum_{s \in \mathcal{S}} \sum_{a \in \mathcal{A}} x(s,a)\, r(s,a),$$

such that $x(s,a) \geq 0$ and for all $s' \in \mathcal{S}$

$$\sum_{a \in \mathcal{A}} x(s',a) - \sum_{s \in \mathcal{S}} \sum_{a \in \mathcal{A}} \gamma p(s' \mid s,a)\, x(s,a) = y(s')$$

with $\boldsymbol{y} \in \mathbb{\Delta}(\mathcal{S})$.

Linear programming

Theorem 19

For any policy π,

$$x_\pi(s,a) = \mathbb{E}^\pi \left\{ \sum_{t \geq 0} \gamma^t \, \mathbb{I}\left\{ s_t = s, a_t = a \mid s_0 \sim \boldsymbol{y} \right\} \right\}$$

is a feasible solution to the dual problem. On the other hand, if $\boldsymbol{x}$ is a feasible solution to the dual problem, then $\sum_a x(s,a) > 0$. Defining a respective randomized policy

$$\pi_{\boldsymbol{x}}(a \mid s) = \frac{x(s,a)}{\sum_{a' \in \mathcal{A}} x(s,a')},$$

$\boldsymbol{x}_{\pi_{\boldsymbol{x}}} = \boldsymbol{x}$ is a feasible solution of the dual program.

Complexity

- VI: $\tilde{O}(|\mathcal{S}|^2|\mathcal{A}|L(1-\gamma)^{-1})$, where L is the total number of bits to represent the MDP.
- PI: $\tilde{O}(|\mathcal{S}|^2|\mathcal{A}|(1-\gamma)^{-1})$, again omitting logarithmic factors.
- LP: $O(|\mathcal{S}|^6 L)$.

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